

Insurance Claim Risk Modeling: Detecting Anomalies in Healthcare Billing

1. Introduction

Healthcare costs in the United States have reached historic highs, with insurance companies routinely processing hundreds of thousands of claims each year. Amid this volume, some claims deviate sharply from typical patterns, whether due to fraud, billing errors, or legitimate but rare medical needs. Detecting these high-risk claims is both a financial and ethical imperative. Fraudulent or excessive billing not only inflates premiums for honest consumers but also undermines trust in the healthcare system.

Traditionally, insurance fraud detection has relied on predefined business rules, whistleblower reports, or manual audits. However, these methods struggle with adaptability and scale. Fraud tactics evolve quickly, and what once looked suspicious may become commonplace, or vice versa. Furthermore, many high-risk claims are not fraudulent in the legal sense but are still financially hazardous or unusual.

Machine learning, particularly anomaly detection techniques, offers a powerful alternative. These models can sift through large volumes of claims and surface outliers cases that deviate meaningfully from the learned norms. By combining several unsupervised techniques with a downstream classification model, we can not only find these anomalies but also prioritize which ones require attention based on a calculated risk score.

This project aims to develop a risk-scoring model to identify potentially high-risk, anomalous, or fraudulent insurance claims in healthcare settings using patient and billing data. The ultimate aim is to create a risk modeling pipeline that helps insurance providers flag claims that warrant closer inspection. While fraud detection is one motivation, the broader goal is to identify any claims that significantly deviate from normal behavior in terms of cost, frequency, or provider behavior.

2. Data Explanation and Data Preparation

To develop and test the model, a synthetic dataset from Kaggle was used. Though artificial, the dataset simulates realistic healthcare scenarios, including both patient and provider attributes. Each record corresponds to a single claim and includes information such as patient demographics (age, gender, blood type), clinical data (diagnosis, length of stay, medications), and financial data (total billing amount, doctor and hospital identifiers).

One of the key steps early in the project was extensive data cleaning and transformation. Missing values were addressed using simple statistical imputation, mean values for numerical fields and the most frequent category for categorical fields. Given the importance of financial behavior in risk detection, much of the feature engineering centered around quantifying how a claim compares to historical averages. For example, new features such as "billing per day," "doctor visit frequency," and "average billing per provider" were calculated. These metrics helped normalize the data across doctors and hospitals with differing patient loads.

To prepare the dataset for modeling, numerical features were scaled using MinMaxScaler, and categorical features like hospital and doctor IDs were frequency-encoded. This allowed the models to focus on patterns of occurrence rather than the raw categorical values.

3. Methods

A two-tiered modeling approach was adopted. First, multiple unsupervised models were used to assign anomaly scores to each claim. These models included Isolation Forest, Local Outlier Factor (LOF), KMeans clustering, DBSCAN, and Autoencoders.

Isolation Forest works by randomly selecting features and splitting values to isolate data points. The fewer splits required to isolate a point, the more anomalous it is considered. LOF, in contrast, compares the local density of a data point to that of its neighbors; outliers are those with significantly lower density. KMeans and DBSCAN helped identify clusters of normal behavior, with small or low-density clusters likely representing anomalies. Autoencoders, a neural network model, were particularly effective, they compress input data into a low-dimensional space and then attempt to reconstruct it. The greater the reconstruction error, the more likely the claim is to be an outlier.

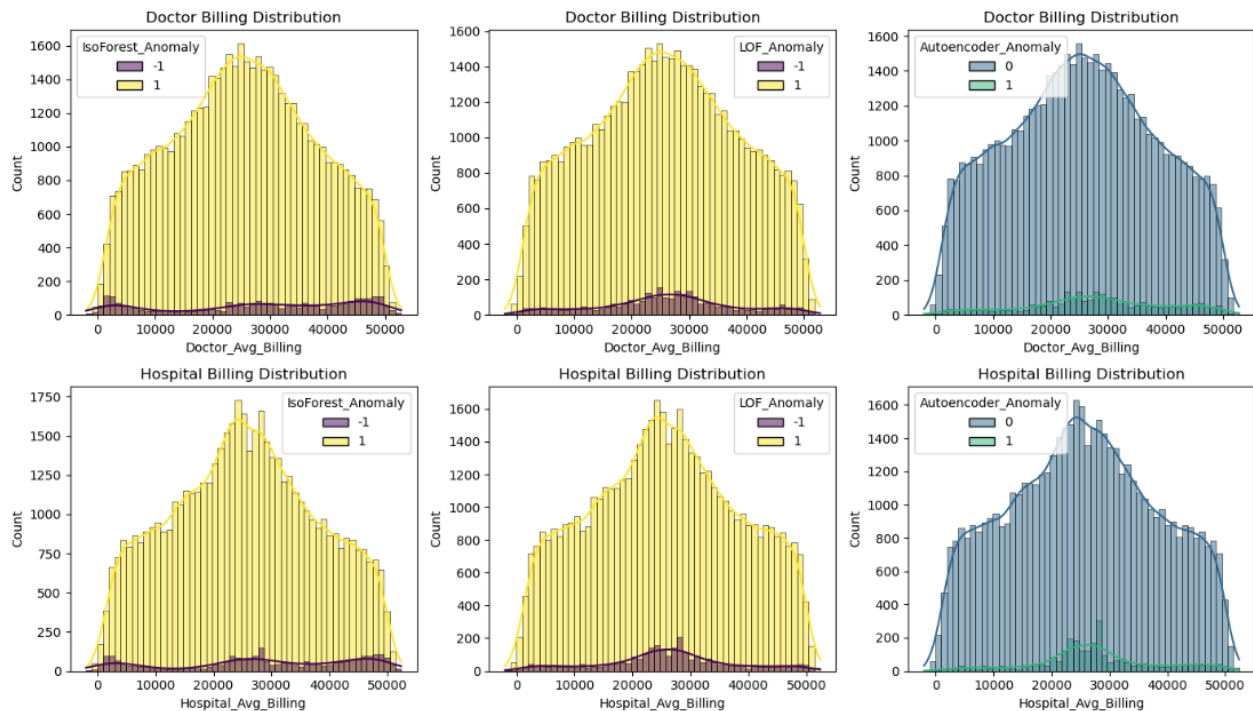
Each method produced a continuous score indicating the likelihood of a claim being anomalous. These scores were combined using a weighted average to form a final "risk score." A threshold of 0.9 was established based on exploratory analysis; claims above this score were considered high risk.

Once the risk labels were established, a supervised learning phase was introduced using Logistic Regression and Random Forest classifiers. To address class imbalance (since high-risk claims were rare), SMOTE (Synthetic Minority Oversampling Technique) was applied. This step helped the classifier learn more effectively from the few anomalous cases we had. Then the performance was the model was evaluated using precision, recall and F1-score.

4. Analysis

a. Anomaly Detection Results:

The application of three distinct anomaly detection algorithms – Isolation Forest, Local Outlier Factor (LOF), and an Autoencoder – yielded complementary perspectives on potential healthcare claim anomalies.



Graph 1: Doctors and Hospital bill distribution

b. Doctor Billing Distribution:

As it can be seen in the above graph 1, the Isolation Forest results (top-left) show that most doctor billings fall between \$10,000 and \$40,000. Anomalies (marked in purple) mostly appeared at the extremes very low billings under \$5,000 and very high billings over \$45,000 which makes sense, as those amounts are quite rare.

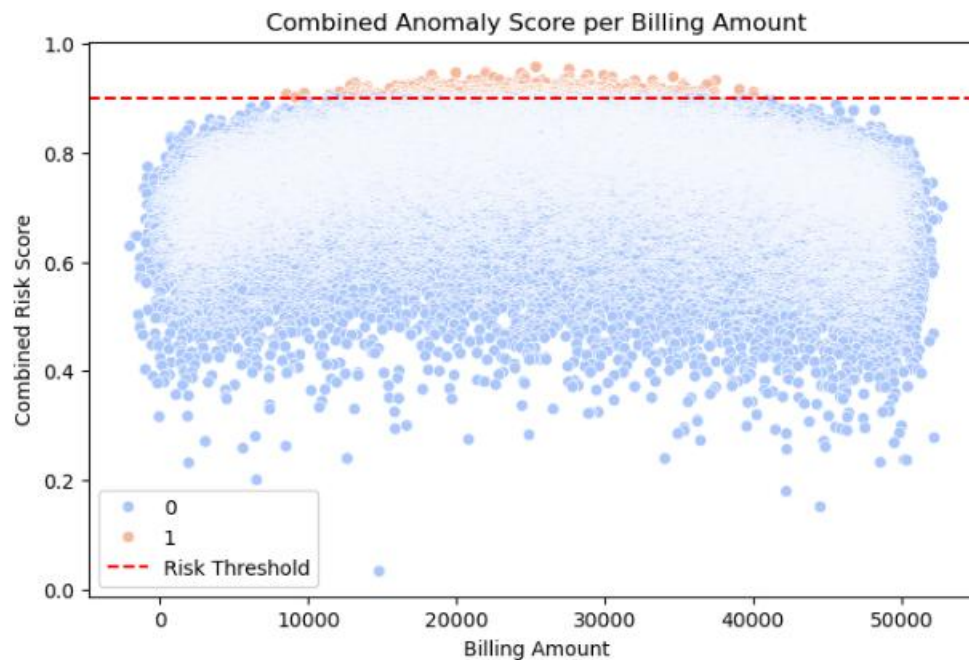
The LOF results (top-middle) were similar, flagging both very low and very high billings as anomalies. However, LOF seemed a bit more selective, catching fewer anomalies overall, especially around the edges.

The Autoencoder (top-right) revealed a slightly different behavior. It mainly flagged high billing amounts as anomalies, while catching fewer low-billing cases compared to Isolation Forest and LOF. This suggests the Autoencoder focused more on inconsistencies at the higher end of the billing spectrum.

c. Hospital Billing Distribution:

The Isolation Forest results (bottom-left) show a familiar pattern, with most hospital billings falling between \$10,000 and \$40,000. Anomalies mainly appeared at the extremes, very low and very high billing amounts, similar to what we saw with doctor billing. LOF (bottom-middle) also followed a similar trend but was a bit more conservative, identifying slightly fewer anomalies at both the low and high ends. On the other hand, the Autoencoder (bottom-right) stood out by flagging more high-billing hospitals as anomalies, while being less sensitive to low-billing cases compared to Isolation Forest.

To better understand the distribution and behavior of anomaly scores in relation to billing amounts, a scatter plot, shown in graph 2, also was created to visualize the combined anomaly scores against the total billing amount for each claim. This plot provides clear insight into how the model differentiates between normal and potentially high-risk claims.



Graph 2: Combined anomaly score

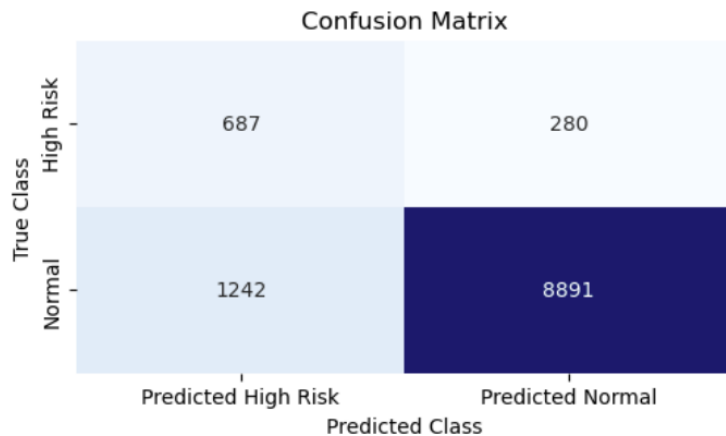
From the plot, it is evident that most claims, regardless of billing amount, cluster well below the risk threshold. This suggests that the majority of the data conforms to expected patterns. However, a distinct band of points rises above the threshold, particularly in the mid-to-high billing ranges (between \$10,000 and \$40,000), where the model has flagged several claims as potentially anomalous. These claims have unusually high risk scores despite billing amounts that may not be the absolute highest, indicating that the model is capturing more nuanced patterns beyond just the raw cost.

Interestingly, the model does not simply flag the most expensive claims as anomalous. While some very high billing amounts do appear near the top, there are also many high-cost claims with moderate or low risk scores. This shows that billing amount alone is not the sole driver of anomaly detection, contextual variables such as patient demographics, length of stay, or provider behavior likely contributed to the combined risk score.

This visualization supports the strength of the ensemble approach in identifying high-risk claims across a wide range of billing levels. It demonstrates that the model is capable of finding anomalies based on complex interactions in the data rather than relying on any single feature.

d. Model Performance

Following the anomaly detection phase, a supervised classification model was developed to help identify high-risk insurance claims with greater automation and precision. The best-performing model demonstrated a training score of 0.867, correctly predicting approximately 86.7% of the training set. The test accuracy was 0.86, which closely mirrors the training performance, suggesting that the model generalizes well to unseen data and is not significantly overfitting.



Training Data Score: 0.8676351351351351					
Accuracy of the best model on test dataset: 0.86					
	precision	recall	f1-score	support	
0	0.36	0.71	0.47	967	
1	0.97	0.88	0.92	10133	
accuracy			0.86	11100	
macro avg	0.66	0.79	0.70	11100	
weighted avg	0.92	0.86	0.88	11100	

Graph 3: Confusion matrix

For Class 0, representing high-risk or potentially fraudulent claims, the model achieved a precision of 0.36, meaning just over one-third of the claims predicted as high-risk were actually true positives. However, it had a relatively strong recall of 0.71, correctly identifying 71% of all actual high-risk claims. The F1-score for this class was 0.47, indicating that while the model is effective at catching high-risk claims, it still returns a fair number of false positives an expected trade-off when recall is prioritized.

In contrast, Class 1, which includes normal claims, showed excellent performance. The model achieved a precision of 0.97, a recall of 0.88, and an F1-score of 0.92, demonstrating that it can very reliably identify and correctly classify standard, non-fraudulent claims.

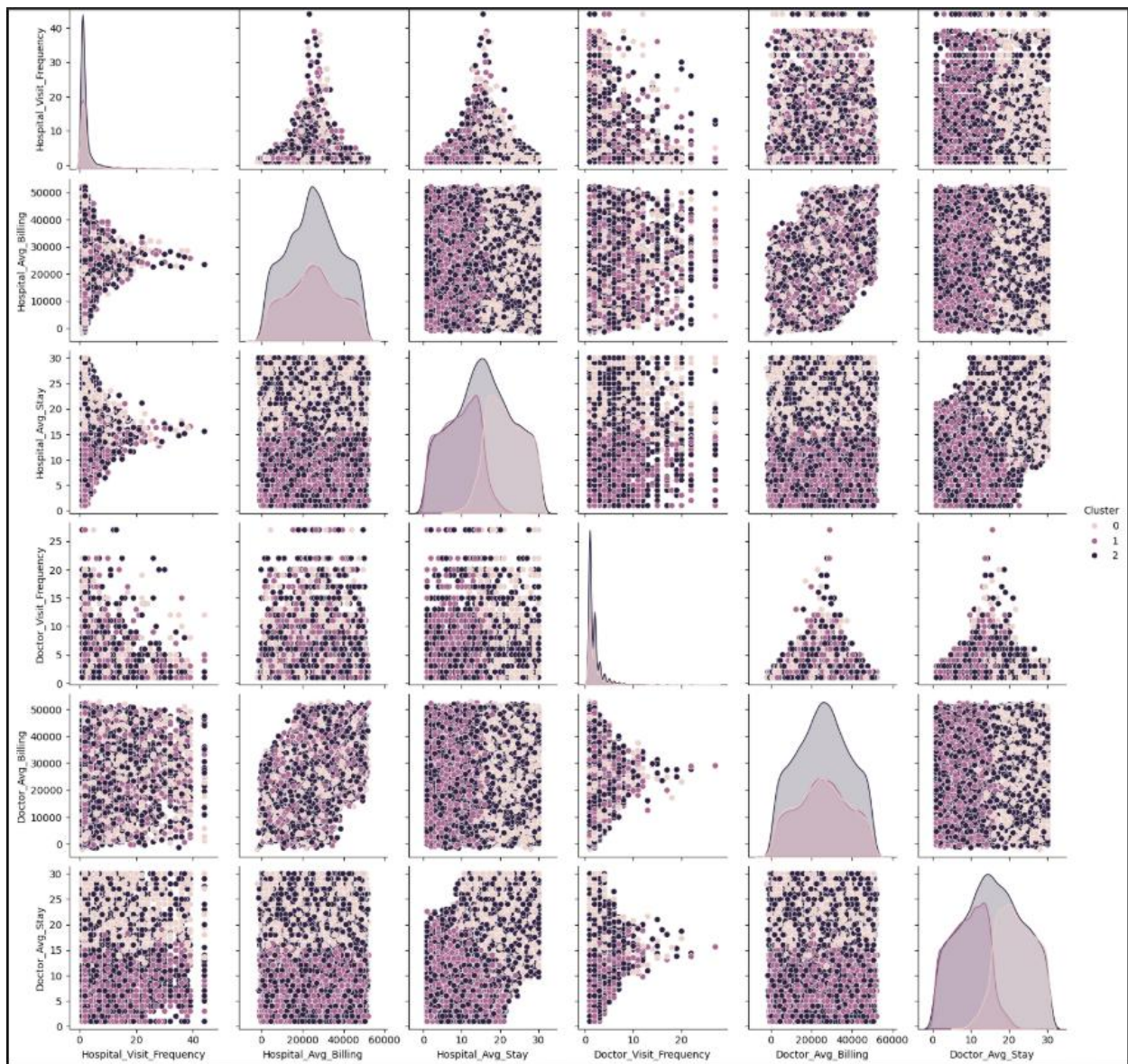
Looking at the aggregated results, the macro average precision and recall were 0.66 and 0.79, respectively. This average gives equal weight to both classes and reflects the difficulty the model has in predicting the minority (fraudulent) class. Meanwhile, the weighted average F1-score was 0.88, which is dominated by the majority class due to class imbalance but still reflects strong overall performance.

These metrics are further validated by the confusion matrix, which shows a substantial number of true positives for Class 0 but also a moderate amount of false positives cases where legitimate claims were flagged as risky. This is a tolerable outcome in a scenario where the cost of missing fraudulent claims is higher than the cost of investigating a few extra legitimate ones.

Overall, the model is progressing in the right direction for a practical fraud detection system. With a 71% recall for high-risk claims, it effectively captures most suspicious cases. The next step in refinement would be to improve precision for Class 0, so that fewer false alarms are raised, enhancing the model's operational usefulness without compromising its protective strength.

e. Pairplot Graph

The pair plot provided valuable insights into the univariate distributions of key features and the bivariate relationships between them, colored by potential patient clusters.



Univariate patterns revealed strong right-skewed distributions for features like Hospital Visit Frequency and Doctor Visit Frequency, indicating that while most patients had limited interactions with healthcare providers, a small subset exhibited frequent usage often aligning with billing anomalies. Similarly, Hospital and Doctor Avg Billing showed a long-tail distribution where a few patients or providers generated abnormally high charges, often flagged by anomaly detection algorithms. In contrast, features like Hospital Avg Stay and Doctor Avg Stay followed more normal distributions, suggesting more consistent behavior, though extreme values here could still represent meaningful outliers.

The bivariate relationships added further depth, with several clusters emerging that combined high doctor and hospital billing. These clusters were especially prone to being flagged as anomalous, hinting at potentially high-risk or fraudulent cases. The visualization helped confirm that flagged anomalies weren't just statistical noise but often came from meaningful groupings in the data. Meanwhile, anomalies found outside of clusters may represent isolated cases that, while less common, still warrant attention due to their deviation from expected patterns.

Bringing this all together, the insights from the pair plot strengthened the overall anomaly detection process. Isolation Forest helped surface broad, unusual billing behaviors, LOF honed in on subtle but locally rare deviations, and the Autoencoder captured more complex inconsistencies involving combinations of features. Together with the observed distributions and clusters, the model ensemble not only flagged potential anomalies but also provided a deeper, interpretable understanding of why certain claims stood out, enabling more targeted and informed follow-up analysis.

5. Conclusion and Recommendations

This study successfully applied a hybrid anomaly detection and classification approach to identify high-risk insurance claims, using a combination of Isolation Forest, Local Outlier Factor, and Autoencoder models. These anomaly scores were integrated into a final risk prediction model Logistic Regression with SMOTE and GridSearchCV optimization. The final model demonstrated strong performance, with a training score of 0.867 and a test accuracy of 0.86, indicating consistent generalization without signs of overfitting. While the overall accuracy is high, a more nuanced view of class performance revealed that the model is particularly effective in detecting the majority class (normal claims) with 97% precision and 88% recall, but remains moderately challenged in predicting the minority class (high-risk claims), with 36% precision and 71% recall.

Visual analysis using anomaly scoring versus billing amount highlighted how high-risk claims typically cluster above a defined threshold. This clear separation, particularly among claims with higher billing amounts, confirms that billing-related anomalies are a strong indicator of potential fraud or abnormal behavior. The pair plot further enriched this understanding by showing distinct patterns in patient visit frequency, billing distributions, and length of stay. These features showed right-skewed distributions with long tails that aligned well with detected anomalies, especially those flagged by the Autoencoder, which is sensitive to complex, multivariate inconsistencies. Clusters with high hospital and doctor billing emerged as critical areas for further investigation, reinforcing that financial outliers are strong predictors of claim risk.

Based on these findings, it is recommended that the anomaly detection pipeline be integrated as a pre-screening tool in the claims evaluation workflow. The current model's high recall for high-risk claims is particularly valuable for minimizing missed fraud cases. However, the relatively low precision suggests that a portion of flagged cases may be false positives, requiring further refinement to reduce unnecessary manual reviews. Future work should focus on enhancing model precision through advanced ensemble methods or cost-sensitive learning. Expanding the feature set to include temporal trends, claim histories, and provider-level metadata could also significantly improve the model's discriminatory power.

6. Assumptions

The dataset reflected a relatively balanced distribution of patient health conditions, ensuring a comprehensive representation across different categories. Additionally, both hospitals and doctors had ample historical records available, allowing for the development of reliable and accurate profiles. This strong data foundation helped support the overall integrity and effectiveness of the analysis.

7. Limitations

Since the dataset was synthetic, there were some important limitations to consider. Without ground truth labels, model evaluation had to rely heavily on assumption-based validation, often requiring manual review to confirm the accuracy of the findings. Additionally, the dataset did not explicitly model skewness in the severity of medical conditions, which could limit the model's ability to capture more complex real-world patterns. These limitations highlight the importance of cautious interpretation and continuous model monitoring when working with synthetic data.

8. Challenges

Despite the promising results, there were several limitations. The dataset was synthetic, which means that while it mimics real-world structures, it may lack the unpredictability and diversity found in actual healthcare systems. Another challenge was choosing an appropriate risk score threshold. An overly aggressive threshold would flag too many normal claims, while a conservative one might miss subtle fraud. The current threshold was chosen based on visual separability, but a more rigorous method using domain feedback would be ideal.

The class imbalance also proved tricky, requiring careful balancing through SMOTE and repeated training/testing to ensure generalization.

9. Future Uses/Additional Applications

Future work should focus on enhancing model precision through advanced ensemble methods or cost-sensitive learning. Additionally, cross-linking patient claims could help detect patterns indicative of organized fraud rings, further strengthening fraud detection efforts.

12. Implementation Plan

Once the model works well in the real world, we'll move on to implementation. The models will be added to the claims management systems, with web dashboards to monitor anomalies in real time. Audit teams will be trained to understand the results, and a system will be put in place to automatically update the models as needed, ensuring they stay accurate and flexible.

13. Ethical Assessment

To ensure the system runs ethically, we need to keep a few key things in mind. First, fairness is essential, any flagged claims should be reviewed by humans to prevent wrongful denials. Privacy is another big concern, so strong data anonymization techniques must be in place to protect sensitive health information. Transparency is also important, with clear, understandable criteria for detecting anomalies that can be easily shared with regulators and policyholders. Lastly, regular audits are needed to ensure the system isn't unintentionally biased against certain groups, providers, or regions.

14. References

- Kuhn, M., & Johnson, K. (2013). *Applied Predictive Modeling*. Springer.
- Articles on healthcare claim fraud detection and medical anomaly detection from journals like *Health Affairs* and *Journal of Biomedical Informatics*
- [5 Essential Stats on Correlation in Insurance Risk Models](#)